Joseph Walton

Python Engineer | AI Engineer | LLM & RAG Specialist

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Professional Summary

Python and AI engineer with over 10 years of software engineering and machine learning experience, including delivery of production LLM, RAG, NLP and agent-based solutions. Proven contract delivery experience taking AI products from evaluation and architecture through deployment, observability and continuous improvement. Combines hands-on engineering with technical leadership, delivering measurable outcomes including approximately £2M in annual savings, £10k in monthly cloud savings and production deployment cycles of under one day.

Core Skills

- **Python Engineering:** Python, SQL, Polars, Pandas, NumPy, PySpark, Azure Functions
- **LLMs, RAG & Agents:** Azure OpenAI, Azure AI Foundry, LangChain, LangGraph, Hugging Face, RAG, prompt and agent evaluation
- **Search & NLP:** Elasticsearch, vector search, HNSW, BERT, FastText, semantic search, recommendation systems
- **Production AI & MLOps:** MLflow, Azure ML Pipelines, Vertex AI Pipelines, CI/CD, observability, evaluation, Docker
- **Cloud Platforms:** Microsoft Azure, Google Cloud Platform, AWS
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn

Professional Experience

IBM

Lead Data Scientist — Contract
June 2025 - Present
Delivering production AI solutions for central government.

- Rescued underperforming AI projects by introducing evidence-based evaluation and observability practices.
- Designed and delivered a Python-based data processing agent using Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Developed a Claude Skill to streamline human-in-the-loop data ingestion for agent evaluations, saving approximately one hour per evaluation case.
- Re-engineered a data matching algorithm from Apache Spark to Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

Tech Stack: Python, Polars, Azure AI Foundry, Azure Functions, LangGraph

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Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developed and operationalised AI and NLP products in a government environment.

- Built a production RAG agent that used unstructured text to help users identify relevant Goods and Services terms.
- Developed generative AI applications within a service-oriented architecture using Python, LangChain, Azure OpenAI and Azure Functions.
- Implemented and tuned Elasticsearch HNSW vector search, balancing throughput and recall through systematic evaluation.
- Improved search and recommendation relevance metrics by 20% through data-driven experimentation.
- Established an end-to-end MLOps strategy across multiple environments, reducing model deployment time to under one day.
- Architected Python ETL pipelines in Polars to replace Azure Databricks workloads, reducing compute costs by 50% and enabling zero-downtime deployments.
- Conducted red-team exercises to assess LLM robustness and improve resilience against adversarial attacks.
- Defined an AI product roadmap and communicated delivery approaches to technical and non-technical stakeholders across international IP offices.

Tech Stack: Python, Azure OpenAI, LangChain, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

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Lead Machine Learning Engineer

February 2020 - November 2021

Led cross-functional delivery of production machine learning products using large-scale text and image data.

- Designed and implemented an MLOps framework on Google Cloud Platform, orchestrating training, evaluation, approval and deployment with Vertex AI Pipelines and reducing time to market to under one day.
- Built Python and Spark pipelines for model training, batch inference and recommendation workloads.
- Developed and deployed NLP and computer vision models, including a multilingual BERT model detecting IP infringements across more than 90 languages.
- Delivered a recommendation engine combining text and image similarity to improve infringement detection.
- Mentored a software engineer through a transition into machine learning engineering.

Tech Stack: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Designed and built a production NLP system for predicting IP infringement in a secure, containerised environment, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Built and deployed an image similarity model that helped secure a major client contract representing 10% of annual recurring revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

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Office for National Statistics

Machine Learning Engineer
September 2017 - November 2018

- Developed Python machine learning models and scalable PySpark pipelines to link and process national administrative datasets.
- Built backend components for the UK Census 2021 data collection system within a cross-functional engineering team.

Tech Stack: Python, PySpark, Cloudera

Earlier Experience

Barclays — Data Scientist (Secondment) | *September 2018 - October 2018*
Developed regional economic indicators from payments data following success in the ONS x Barclays Data Hackathon.

Purple Secure Systems — Software Engineer | *August 2014 - October 2016*
Developed Python, Java and PySpark big-data processing systems for government and defence clients.

Education

MSc Applied Mathematics, University of Bath | *2016 - 2017*
BSc Physics, University of Bath | *2010 - 2013*

Award

Winner, ONS x Barclays Data Hackathon — Used payments data to create more granular and timely estimates for Gross Value Added.